GMS STANDALONE

FOR

TANZANIA POLICE FORCE

(TPF)

Prepared by

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Integrity and Innovation for Quality Public Service Delivery

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1. Introduction

The purpose of this document is to specify the overall TPF Mailing System requirements that will govern the development/acquisition, installation and configuration of the new mailing system. The document will also establish initial security, training, capacity and system architecture requirements, as well as, system acceptance criteria agreed upon by the TPF Management and e-GA.

1.1 Scope

The requirements document will outline the functional, performance, security, high level design and other system requirements identified for the proposed Standalone Mailing System for TPF

2. Current TPF Mailing System Summary

Currently, TPF they use shared Mailing System and in use under domain ***tpf.mil.tz.***
The plan is to implement TPF in-house email system that can be able to handle more than 1500 users for email communication. To start with, 1 physical/virtual server with minimum of 5TB storage, 64 CPU cores and 128GB memory (ROM) as Mailbox and 7 virtual servers with a minimum of 100GB storage will be used to setup the system

2.1 Application Setup

For this setup, users can access emails through web-mail or Desktop Mail client (Outlook) and both incoming and outgoing mails can be set in a separate Server preferably Virtual Machine. i.e.

1. Web-mail [mail.domain.go.tz] →Mail-Server ↔Incoming/Outgoing Mails
2. Mail-TPF[Outlook]→Mail-Server[POP3&SMTP] ↔Incoming/Outgoing Mails

3. TTCL Mailing System Computing and Storage specifications.

The table below highlights some of the suggested TPF Mailing System Computing and Storage specifications.

S/N	Resources	Location	Quantity	Remarks
1	Server 1; (or equivalent specs) (x2) Intel® Xeon® E5-26XX (min. 6 core, 2.4 - 4GHz, 64MB, 95W); with hyper threading support, RAM min. 128GB. HDD with min. 5TB	PRIMARY SITE (PR)	1	Rocky/Alma Linux >8.5, with virtualization environment; Any common virtualization technology can be used.
2	Server 1; (or equivalent specs) (x2) Intel® Xeon® E5-26XX (min. 6 core, 2.4 - 4GHz, 64MB, 95W); with hyper threading support, RAM min. 128GB. HDD with min. 5TB	DISASTER RECOVERY SITE(DR)	1	Rocky/Alma Linux >8.5, with virtualization environment; Any common virtualization technology can be used.

3.	Public IP's		10	6 Public IP's for PR and 4 Public IP's for DR are needed
4.	Private IP's		14	8 Private IP's for PR and 6 Private IP's for DR are needed

To achieve the system performance, availability and stability and also be able avoid risk of getting problem in case a specific hardware server is affected, there is a need of adopt server virtualization technology to minimize those risks.

SERVER SPECIFICATIONS ON PR SITE						
S/N	Server	Server Type	Quantity	RAM(GB)	Storage (GB)	OS to be Deployed
1.	Hypervisor	Physical	1	128	5000	Any Virtualization
2.	MAILBOX	Physical/VM	1	64	3500	Rocky/Alma Linux 8.6 or above
3.	WEB	VM	1	8	100	Rocky/Alma Linux 8.6 or above
4.	FILTER	VM	1	8	100	Rocky/Alma Linux 8.6 or above
5.	ADMIN	VM	1	8	100	Rocky/Alma Linux 8.6 or above
6.	MX	VM	2	8	100	Rocky/Alma Linux 8.6 or above
7.	SMTP	VM	2	8	100	Rocky/Alma Linux 8.6 or above

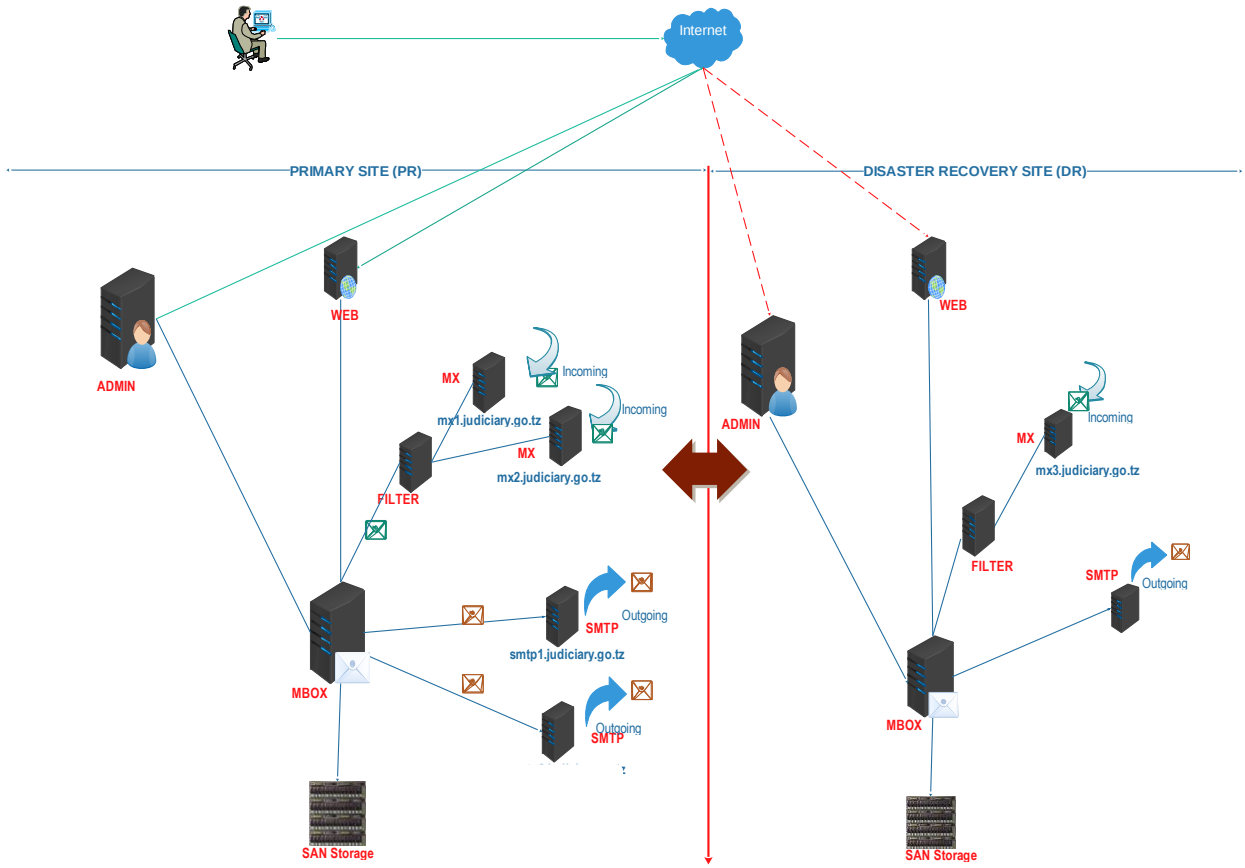
Note: Similar resources or minimum recommendations of the server specifications will be deployed on Disaster Recovery site.

4. General Mailing System Requirements

The requirements described below were the result of discussion made with various mail system users, and analyzed the standards, and possible solutions;

i. Mail Server which will be a collaborative tool to TPF for e-mail, calendar, task, messaging
ii. Full synchronization of emails, calendar, contacts, tasks and notes to all computing device platforms.
iii. Provide centralized automatic of all archiving mails.
iv. Option to fully integrated with outlook
v. Share contacts, calendars, tasks and meeting information including booking and allocation of meeting resources such as meeting room, projector for proper organization of meetings
vi. Provide a common signature with TPF logo for all users.
vii. Provide webmail access with unlimited mails handling
viii. Secure and reliable mail server
ix. Statistics for users logged in, emails sent/received per day,
x. Ability to extract or export emails users.
xi. Ability to show users status i.e. enabled or disabled and account activities.
xii. Ability to provide reasons for emails rejection/bouncing.
xiii. Restrict only legitimate users to send emails and able to track senders to minimize fraud.

5. Proposed TPF Mail System Architecture



5.1 Primary(PR) Site, TPF HQ

The PR Site will be stationed at TPF HQ Data Centre or any other place suggested by TPF. All emails will be retrieved and stored locally at the PR Site. Proper power backup and network connectivity are critical to ensure system's availability.

5.2 Disaster Recovery (DR) Site

The Secondary Site used as the DR Site will also be collocated at the place suggested by TPF. The Primary Site will be connected to DR Site via secure VPN tunnel if there is no physical link and replications for mails will be done by rsync unless the storages has capability of storage to storage replication.

5.3 Data Backup

TPF will suggest location to be used for data backup. User data/emails will regularly be backed up so that they can be retrieved in the event that PR and DR data are completely lost beyond recovery or repair.

6. TPF Mailing System Implementation Plan

S/N	System	Location	Activity Breakdown	Duration (Days)
1	TPF Mail System	Primary (PR) Site	✓ OS Images Preparations (Proxmox /VMware & Rocky/Alma Linux 8.6) ✓ Installation, configuration and setup Physical/VM server and OS (Rocky Linux/Alma 8.6) Installation ✓ Installation of the latest packages or software to the server ✓ Replication Setup for	7

			Databases (MySQL &LDAP) ✓ Mailboxes replication setup ✓ Data and user emails migration from the existing instance to the newly deployed Mail System ✓ Mail System Fine tuning enhancement	
2.	TPF Mailing System	Disaster Recovery (DR) Site	✓ OS Images Preparations (Proxmox /VMware & Rocky/Alma Linux 8.6) ✓ Installation, configuration and setup Physical/VM server and OS (Rocky/Alma	7

			Linux 8.6) Installation ✓ Installation of the latest packages or software to the server ✓ Replication Setup for Databases (MySQL &LDAP) ✓ Mailboxes replication setup ✓ Data and user emails migration from the existing instance to the newly deployed Mail System ✓ Mail System Fine tuning enhancement	
3.	Training & Handover		✓ Technical Training of Backend Administration of TPF	5

			Mailing System to TPF System Administrators ✓ Front End Training of TPF end users(***TPF Technical Team can handle this***)	

7. TPF Mailing System Interfaces and Administration

TPF MAIL SYSTEM will have various interfaces for user access and administration.

The interface variations depend on: -

- ✓ Logging on for administration
- ✓ The screen and page layout.
- ✓ The effect on the interface of different administrator privileges.
- ✓ The effect on the interface of different user privileges

All interfaces can only be accessed using a Web browser, preferably Firefox or Chrome, although other browsers such as Internet Explorer should also work.

7.1 TPF Mailing System End User Interface

TPF MAIL SYSTEM user interface let's all the users read their e-mail using their Web browser. They can read, forward, reply and delete their mail from anywhere once they have access to the Internet.

They can also read, send and organize email messages and manage and share calendars, contacts, tasks, notes, files and bookmarks.

More details on the system user interface will be described in ***Mail User Guide.***

7.2 TPF MAIL SYSTEM Administrator Interface

The interface logs the user into the Administration interface. The options displayed after successful login to the user will depend upon their access rights. Logging in as Postmaster will provide full access to all products.

Depending on administrator's level of privilege, the following actions are allowed:

- ✓ Global administrator — has access to all the system areas and the option to administer domain administrator and users.
- ✓ Domain administrator — in a multiple domain system, looks after one domain, adding users, creating domain profiles, etc. Has access to his user's area only.
- ✓ Webmail Administrator – Has administrator privilege to perform some functionalities in webmail interface.
- ✓ User — cannot perform any management/administrative tasks and he's unable to login into the administrator interface. So just has access to user interface

Big Note:→ The postmaster account has system, domain and logs administrative permissions.

Access to actions is controlled in three ways:

- ✓ If a user is not permitted to perform a complete set of actions, the panel is not visible.
- ✓ If a user has access to just a subset of pages under a button, the tabs for the remaining pages are greyed out.
- ✓ If a user is not permitted to perform any administrative task, he's not allowed to login into administrator interface

More details on the System administrator interface will be described in ***Mail Administrator Guide.***

8. TPF Mail System Day-to-Day Administration and Management

Mail System day-to-day management involves all administrators. It describes the main tasks to be carry out on a regular basis. Operations like customizing the user inter- face and upgrading the system is not covered here.

The following areas are among the day-to-day activities:

- ✓ Accounts overview – individual & special accounts
- ✓ Managing accounts - includes adding multiple accounts, Enable and Disable accounts.
- ✓ Account attributes - aliases, forwarding accounts and "moved" messages.
- ✓ Setting up auto responders.
- ✓ Groups overview - how to create groups, add members and post to a group.
- ✓ Managing Calendars, Tasks, notes, tasks and folders.
- ✓ Mailing all users in a domain.
- ✓ Maintaining logs - specifying log levels, configuring log handling, deleting/compressing and e-mailing a log, searching logs.
- ✓ Blacklist & Whitelist accounts.
- ✓ Etc.

Big Note:→ More details on the Mail System Day-to-day Management will be described in both System Administrator and User Guide